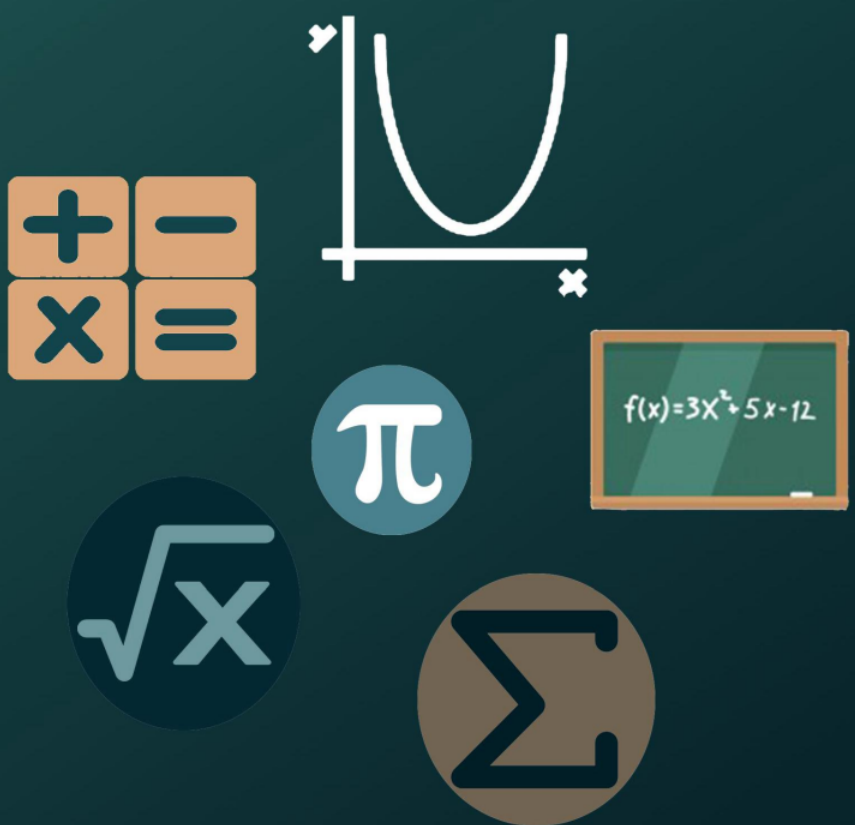
MATHEMATICS

Compound Angle

(Trigonometric Ratios & Identities)



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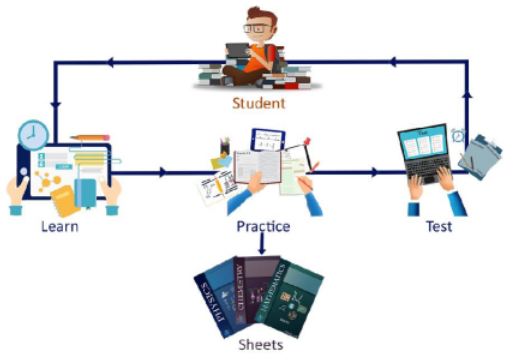
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Trigonometry	1-2	12-14		
2	Trigonometric Ratios	3-8	2, 11, 15-17	1-2	19-20
3	Real Definition of Trigonometric Ratios				
4	Trigonometric Ratios of Allied Angles	9-15	18-21	3	
5	Trigonometric Graphs				
6	Trigonometric Ratios of Compound Angles	16-18	22		
7	Transformation Formulae - Sum to Product	19-20	23-25	4	14-15
8	Transformation Formulae - Product to Sum	21-23	26-28		
9	Tangent Formula	24	29		1-2
10	Trigonometric Ratios of Multiple & Submultiple Angles	25-30	2-3, 6-7, 30-31	5-7	3-7, 16-17, 21-24
11	Trigonometric Ratios for some Standard Angles	31	4, 32-33	8	25-26
12	Trigonometric Ratios of the Sum of more than Two Angles	32-35			
13	Trigonometric Identities in a Triangle (Conditional Identities)		34-37		8
14	Application of Trigonometry in Maximising & Minimising - Type I		38-39		
15	Application of Trigonometry in Maximising & Minimising - Type II, III		5, 40-41		
16	Application of Trigonometry in Maximising & Minimising - Type IV, V & VI		42-44	9-13	
17	Summation of Trigonometric Series	36-40			9, 27
18	Continued Product of Sine & Cosine Series	41	8-10, 45	14-16	10
19	After Complete Chapter	42-43		17-18	11-13, 18

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Measurement of Angle

Q1. 75° is equal to -

(A) $\left(\frac{\pi}{9}\right)^C$ (B) $\left(\frac{5\pi}{12}\right)^C$

(C) $\left(\frac{17\pi}{9}\right)^C$ (D) $\left(\frac{16\pi}{9}\right)^C$

Q2. 80^g is equal to -

(A) 73° (B) 75°

(C) 72° (D) None of these

Trigonometric Ratios & Basic Identities

Q3. If $\sin x + \sin^2 x = 1$ then the value of $\cos^2 x + \cos^4 x$ is -

(A) 0 (B) 2

(C) 1 (D) None of these

Q4. If $\sin\theta + \cos\theta = 1$ then $\sin\theta \cos\theta$ =

(A) 0 (B) 1

(C) 2 (D) $\frac{1}{2}$

Q5. $2(\sin^6\theta + \cos^6\theta) - 3(\sin^4\theta + \cos^4\theta) + 1$ is equal to-

(A) 2 (B) 0

(C) 4 (D) 6

Q6. If $x \sin 45° \cos^2 60° = \frac{\tan^2 60° \operatorname{cosec} 30°}{\sec 45° \cot^2 30°}$, then x =

(A) 2 (B) 4

(C) 8 (D) 16

Q7. If $\sin\theta = \frac{24}{25}$ and θ lies in the second quadrant then $\sec\theta + \tan\theta$ =

(A) –3 (B) –5 (C) –7 (D) –9

Q8. If $\operatorname{cosec} A + \cot A = \frac{11}{2}$ then tan A is equal to-

(A) $\frac{22}{117}$ (B) $\frac{2}{11}$

(C) $\frac{117}{22}$ (D) $\frac{44}{117}$

Trigonometric Ratios of Allied Angles

Q9. $\tan A + \cot(180° + A) + \cot(90° + A) + \cot(360° - A)$ =

(A) 0 (B) 2 tan A

(C) 2 cot A (D) 2 (tan A– cot A)

Q10. $\sin(\pi + \theta)\sin(\pi - \theta)\operatorname{cosec}^2\theta$ =

(A) 1 (B) –1

(C) $\sin\theta$ (D) $-\sin\theta$

Q11. $\cos(-1710°)$

(A) $\frac{1}{\sqrt{2}}$ (B) $-\frac{1}{\sqrt{2}}$

(C) 1 (D) 0

Q12. $\cos 510° \cos 330° + \sin 390° \cos 120°$ =

(A) 2 (B) –1 (C) 0 (D) $\frac{1}{\sqrt{2}}$

Q13. The expression

$$\frac{\tan\left(x-\frac{\pi}{2}\right).\cos\left(\frac{3\pi}{2}+x\right)-\sin^3\left(\frac{7\pi}{2}-x\right)}{\cos\left(x-\frac{\pi}{2}\right).\tan\left(\frac{3\pi}{2}+x\right)}$$

simplifies to -

(A) $(1 + \cos^2 x)$ (B) $\sin^2 x$

(C) $-(1 + \cos^2 x)$ (D) $\cos^2 x$

Q14. Exact value of $\cos^2 73° + \cos^2 47° - \sin^2 43° + \sin^2 107°$ is equal to -

(A) 1/2 (B) 3/4

(C) 1 (D) none

Q15. Value of $\sin^2\frac{\pi}{8}+\sin^2\frac{3\pi}{8}+\sin^2\frac{5\pi}{8}+\sin^2\frac{7\pi}{8}$ is :-

(A) 2 (B) 1

(C) $1\frac{1}{8}$ (D) $2\frac{1}{8}$

Trigonometric Ratio of Compound Angles

Q16. If $\cot\alpha\ \cot\beta = 2$, then $\frac{\cos(\alpha+\beta)}{\cos(\alpha-\beta)} =$

(A) $\frac{1}{3}$ (B) $-\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $-\frac{1}{2}$

Q17. $\frac{\sin(A-C)+2\sin A+\sin(A+C)}{\sin(B-C)+2\sin B+\sin(B+C)}$ is equal to-

(A) tan A (B) $\frac{\sin A}{\sin B}$

(C) $\frac{\cos A}{\cos B}$ (D) $\frac{\sin C}{\cos B}$

Q18. The value of $\cos^2\frac{\theta-\phi}{2}-\sin^2\frac{\theta+\phi}{2} =$

(A) $\sin\theta\ \cos\phi$ (B) $\sin\theta\ \sin\phi$

(C) $\cos\theta\ \cos\phi$ (D) $\cos\theta\ \sin\phi$

Transformation Formulae : Sum to Product

Q19. $\frac{\cos 4x+\cos 3x+\cos 2x}{\sin 4x+\sin 3x+\sin 2x}$ is equal to-

(A) cot 3x (B) cos 3x

(C) tan 3x (D) None of these

Q20. $(\sin3A + \sin A)\sin A + (\cos3A - \cos A)\cos A = ...$

(A) sinA (B) cosA

(C) 1 (D) 0

Transformation Formulae : Product to Sum

Q21. The value of $2\sin\left(\frac{5\pi}{12}\right)\sin\left(\frac{\pi}{12}\right) =$

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$

(C) $\frac{1}{4}$ (D) $\frac{1}{6}$

Q22. The expression $\frac{\sin 8\theta\cos\theta-\sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta-\sin 3\theta\sin 4\theta}$ is equals-

(A) tan θ (B) tan 2θ

(C) sin 2θ (D) None of these

Q23. The product cot 123°. cot 133° . cot 137° . cot 147°, when simplified is equal to -

(A) – 1 (B) tan 37°

(C) cot 33° (D) 1

Tangent Formula

Q24. If $\tan A = \frac{\alpha}{\alpha+1}$ and $\tan B = \frac{1}{2\alpha+1}$ then A + B is equal to :-

(A) 0 (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Trigonometric Ratios of Multiple & Submultiple Angles

Q25. $1-2\sin^2\left(\frac{\pi}{4}+\theta\right) =$

(A) cos 2θ (B) –cos 2θ

(C) sin 2θ (D) –sin 2θ

Q26. $\frac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta}=$

(A) $\cot\left(\frac{\theta}{2}\right)$ (B) $\sin\left(\frac{\theta}{2}\right)$

(C) $\cos\left(\frac{\theta}{2}\right)$ (D) $\tan\left(\frac{\theta}{2}\right)$

Q27. $\cos 22\frac{1}{2}^{\circ}$ is equal to :-

(A) $\frac{1}{2}\sqrt{2+\sqrt{2}}$ (B) $\frac{1}{2}\sqrt{2-\sqrt{2}}$

(C) $\sqrt{2}-1$ (D) $\sqrt{2}+1$

Q28. $\frac{3\cos\theta+\cos 3\theta}{3\sin\theta-\sin 3\theta}$ is equal to-

(A) $1+\cot^2\theta$ (B) $\cot^4\theta$

(C) $\cot^3\theta$ (D) $2\cot\theta$

Q29. Exact value of $\cos 20^\circ + 2\sin^2 55^\circ - \sqrt{2}\sin 65^\circ$ is -

(A) 1 (B) $\frac{1}{\sqrt{2}}$

(C) $\sqrt{2}$ (D) zero

Q30. In ΔABC, $a=3$, $b=4$ and $c=5$, then value of $\sin A + \sin 2B + \sin 3C$ is -

(A) $\frac{39}{25}$ (B) $\frac{12}{25}$

(C) $\frac{14}{25}$ (D) None

Trigonometric Ratio for Some Standard Angles

Q31. Which of the following is rational :-

(A) $\sin 15^\circ$ (B) $\cos 15^\circ$

(C) $\tan 15^\circ$ (D) None

Conditional Identities

Q32. In ΔABC, value of $\cos 2A + \cos 2B + \cos 2C$ equals :-

(A) $-1 - 4\cos A\cos B\cos C$

(B) $-4\cos A\cos B\cos C$

(C) $-1 + 4\cos A\cos B\cos C$

(D) $-1 - 4\sin A\sin B\sin C$

Q33. If $A + B + C = \pi$ then value of expression $\frac{\cos A}{\sin B\sin C}+\frac{\cos B}{\sin C\sin A}+\frac{\cos C}{\sin A\sin B}$ is :-

(A) 1 (B) 2

(C) 3 (D) 4

Q34. If $A + B + C = 180^\circ$ then $\sec A(\cos B\cos C - \sin B\sin C)$ is equal to :-

(A) 0 (B) –1

(C) 1 (D) None

Q35. If $A+B+C=\pi$ then $\frac{\tan A+\tan B+\tan C}{\tan A\tan B\tan C}=$

(A) 0 (B) 2

(C) 1 (D) –1

Application of Trigonometry in Maximising and Minimising

Q36. The greatest and least value of $\sin\theta\cos\theta$ are-

(A) 1,–1 (B) $-\frac{1}{2},-\frac{1}{2}$

(C) $\frac{1}{2},-\frac{1}{2}$ (D) 2,–1

Q37. The minimum value of $\sin\theta+\sqrt{3}\cos\theta$ is-

(A) 2 (B) 0

(C) 1 (D) –2

Q38. The range of $3\cos^2\theta + 5\sin^2\theta$ is :-

(A) [3,4] (B) [4,5]

(C) [3,5] (D) None

Q39. The maximum value of

$$\sin\left(\theta+\frac{\pi}{6}\right)+\cos\left(\theta+\frac{\pi}{6}\right)$$

is attained at $\theta \in (0, \pi/2)$.

(A) $\frac{\pi}{12}$ (B) $\frac{\pi}{6}$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Q40. The least value of $3\sin\theta + 4\cos\theta + 5$ is-

(A) 5 (B) 9

(C) 0 (D) 10

Summation of Trigonometric Series

Q41. The value of $\cos\frac{\pi}{7}+\cos\frac{2\pi}{7}+\cos\frac{3\pi}{7}$

$+\cos\frac{4\pi}{7}+\cos\frac{5\pi}{7}+\cos\frac{6\pi}{7}+\cos\frac{7\pi}{7}$ is -

(A) 1 (B) –1

(C) 1/2 (D) 0

Continued Product of Sine and Cosine Series

Q42. $\cos\frac{\pi}{5}\cos\frac{2\pi}{5}\cos\frac{4\pi}{5}\cos\frac{8\pi}{5} =$

(A) $\frac{1}{16}$ (B) 0

(C) $-\frac{1}{8}$ (D) $-\frac{1}{16}$

Q43. The value of

$$64\sqrt{3}\sin\frac{\pi}{48}\cos\frac{\pi}{48}\cos\frac{\pi}{24}\cos\frac{\pi}{12}\cos\frac{\pi}{6}$$

is -

(A) 8 (B) 6

(C) 4 (D) 12

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If $\sqrt{\frac{1-\sin A}{1+\sin A}}+\frac{\sin A}{\cos A}=\frac{1}{\cos A}$, for all permissible values of A, then A belongs to -

(A) First Quadrant

(B) Second Quadrant

(C) Third Quadrant

(D) Fourth Quadrant

Q2. If $\cot x = \frac{4}{3}$, $x \in \left(\pi, \frac{3\pi}{2}\right)$ and $\tan y = \frac{-5}{12}$, $y \in \left(\frac{3\pi}{2}, 2\pi\right)$ then which of the following is (are) CORRECT?

(A) $\sin(x + y) = \frac{-56}{65}$

(B) $\cos(x - y) = \frac{-33}{65}$

(C) $\sin 3x = \frac{-117}{125}$

(D) $\cos 2y = \frac{119}{169}$

Q3. Which of the following identities wherever defined hold(s) good?

(A) $\cos 3\theta = 4\cos\theta - 3\cos^3\theta$

(B) $\tan\left(\frac{\pi}{4}-\theta\right)=\frac{1-\sin 2\theta}{\cos 2\theta}$

(C) $\tan\theta=\frac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}$

(D) $\sin^3\theta = \sin\theta \sin(60^\circ - \theta) \sin(60^\circ + \theta)$

Q4. If x satisfies $\log_2 x + \log_x 2 = 4$, then $\log_2 x$ can be -

(A) $\tan(\pi/12)$ (B) $\tan(\pi/8)$

(C) $\tan(5\pi/12)$ (D) $\tan(3\pi/8)$

Q5. Let $f(x) = 11 - 8\sin x - 2\cos^2 x$ if the maximum & minimum values of f(x) are denoted by M & m respectively then which of the following is / are correct?

(A) M = 19

(B) m = 3

(C) m = – 3

(D) Range of f(x) is (3, 19]

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 6 to 7)

Let a = cos10°, b = cos50° and c = cos 70°.

Q6. The value of $(a^2 + b^2 + c^2)$ is equal to :-

(A) 2 (B) $\frac{3}{2}$

(C) $\frac{4}{3}$ (D) 1

Q7. The value of $4(a^2 - bc)$ is equal to -

(A) $\frac{8}{3}$ (B) $\frac{5}{2}$

(C) 2 (D) 3

Paragraph # 2 (Ques. 8 to 10)

Continued product $\cos\alpha\cos2\alpha\cos2^2\alpha \ldots\ldots \cos2^{n-1}\alpha$

$$=\begin{cases}\frac{\sin 2^n\alpha}{2^n\sin\alpha}, & \text{if } \alpha \neq n\pi \\ \frac{1}{2^n}, & \text{if } \alpha=\frac{\pi}{2^n+1} \quad \text{i.e. } 2^n\alpha=\pi-\alpha \\ -\frac{1}{2^n}, & \text{if } \alpha=\frac{\pi}{2^n-1} \text{ i.e. } 2^n\alpha=\pi+\alpha\end{cases}$$

Where, $n \in I$ (Integer)

Q8. The value of $\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\cos\frac{6\pi}{7}$ is -

(A) –1/2 (B) 1/2

(C) 1/4 (D) 1/8

Q9. If $\alpha = \frac{\pi}{15}$, then the value of $\prod_{r=1}^{7}\cos r\alpha$ is-

(A) $\frac{1}{128}$ (B) $-\frac{1}{128}$

(C) $\frac{1}{64}$ (D) $\frac{1}{32}$

Q10. The value of $\sin\left(\frac{\pi}{14}\right)\sin\left(\frac{3\pi}{14}\right)\sin\left(\frac{5\pi}{14}\right)\sin\left(\frac{7\pi}{14}\right)\sin\left(\frac{9\pi}{14}\right)\sin\left(\frac{11\pi}{14}\right)\sin\left(\frac{13\pi}{14}\right)$ is-

(A) 1 (B) $\frac{1}{8}$

(C) $\frac{1}{32}$ (D) $\frac{1}{64}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q11. Given : $\frac{\sin\alpha}{a}=\frac{\cos\alpha}{b}=\frac{\tan\alpha}{c}=k$.

Column-I	Column-II
(A) a^2+b^2	(P) $\frac{1}{b^2k^4}$
(B) $a^2+b^2+c^2$	(Q) $\frac{1}{k^2}$
(C) bc	(R) $\frac{1}{ak}$
(D) $\frac{1}{ck}+\frac{ak}{1+bk}$	(S) $\frac{a}{k}$

SUBJECTIVE TYPE

Q12. Consider a square of side 4 cm. Now if an insect runs outside the square at a distance of 1cm from the nearest side of the square. How much distance will it travel in one round. [$\pi = 3.14$]

Q13. Mark the angles and quadrants : 120°, 750°, –300°.

Q14. If the arcs of same length in two circles subtend angles of 60° and 75° at their centres. Find the ratio of their radii.

Q15. Prove that : (cosecA – sinA) (secA –cosA) (tanA + cotA) = 1.

Q16. Prove that : $\frac{\cot A+\tan B}{\cot B+\tan A}=\cot A\tan B$.

Q17. Prove that (sec θ + cosec θ)(sin θ + cos θ) = sec θ cosec θ + 2.

Q18. Find the values of :

(i) $\tan(11\pi)$ (ii) $\tan\left(\frac{11\pi}{6}\right)$

(iii) $\sin\left(\theta-\frac{\pi}{2}\right)$ (iv) $\log_2 \operatorname{cosec}\left(\frac{3\pi}{4}\right)$

Q19. If $\sin\theta=-\frac{1}{2}$ then find the two possible values of θ in (0, 2π).

Q20. If $A=\sin\theta+\sin\left(\frac{\pi}{2}+\theta\right)$,

$B=\cos\theta+\cos\left(\frac{\pi}{2}+\theta\right)$

then prove that $A^2+B^2=2$.

Q21. Find the sign of sin140° + cos140°

Q22. Prove that : $\frac{\tan(A+B)}{\cot(A-B)}=\frac{\sin^2A-\sin^2B}{\cos^2A-\sin^2B}$

Q23. Evaluate $\left(\frac{\sin75°-\sin15°}{\cos75°+\cos15°}\right)^8$

Q24. Evaluate $\frac{(\cos\theta-\cos3\theta)(\sin8\theta+\sin2\theta)}{(\sin5\theta-\sin\theta)(\cos4\theta-\cos6\theta)}$.

Q25. Prove that $\frac{\sin A+\sin B}{\cos A+\cos B}=\tan\frac{A+B}{2}$

Q26. Prove that :

$\cos2\theta\cos\frac{\theta}{2}-\cos3\theta\cos\frac{9\theta}{2}=\sin5\theta\sin\frac{5\theta}{2}$

Q27. Prove that :

$\sin(45°+A)\sin(45°-A)=\frac{1}{2}\cos2A$

Q28. Prove that :

$\frac{2\sin(A-C)\cos C-\sin(A-2C)}{2\sin(B-C)\cos C-\sin(B-2C)}=\frac{\sin A}{\sin B}$

Q29. If A + B = 60°, then find the value of $\left(1+\sqrt{3}\tan A\right)\left(1+\sqrt{3}\tan B\right)$.

Q30. Find the value of $8\sin^3 10° - 6\sin 10°$.

Q31. Find the value of $\text{cosec}\ 10° - \sqrt{3}\ \sec 10°$.

Q32. Evaluate $\sin 78° - \sin 66° - \sin 42° + \sin 6°$.

Q33. Prove that $\sin^2 72° - \sin^2 60° = \frac{\sqrt{5}-1}{8}$.

Q34. In a quadrilateral ABCD.
Prove that $\sin(A + B) + \sin(C + D) = 0$.

Q35. Prove that
$\sin 2A + \sin 2B - \sin 2C = 4\cos A \cos B \sin C$
(where $A + B + C = \pi$).

Q36. Prove that
$\cos 2A + \cos 2B - \cos 2C = 1 - 4\sin A \sin B \cos C$
(where $A + B + C = \pi$).

Q37. Find $\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C}$
(where $A + B + C = \pi$).

Q38. Find the range of following expressions-
(i) $y = \sin\sqrt{x}$ (ii) $y = \sin^2 x$
(iii) $y = \sin 3x$

Q39. Find the range of
$y = \sin^2\left(\frac{15\pi}{8} - 4x\right) - \sin^2\left(\frac{17\pi}{8} - 4x\right)$.

Q40. If the maximum and minimum values of $y = \sin\left(x + \frac{\pi}{6}\right) + 3\cos\left(x - \frac{\pi}{3}\right)$ are A and B. Find the value of A+B.

Q41. Find maximum and minimum value of following expressions-
(i) $y = 5\sin 2x + 12\cos 2x$
(ii) $y = 5\sin(3x + 2) + 12\cos(3x + 2) + 7$

Q42. Find the minimum value of the following expressions-
(i) $9\tan^2\theta + 16\cot^2\theta$
(ii) $9\sec^2\theta + 16\ \text{cosec}^2\theta$

Q43. Find the minimum value of
$y = 4\sin^2 x + 27\ \text{cosec}^2 x$.

Q44. Find minimum value of
$y = 18\ \sec^2 x + 8\cos^2 x$.

Q45. Prove that : $\cos\frac{\pi}{9}\cos\frac{2\pi}{9}\cos\frac{3\pi}{9}\cos\frac{4\pi}{9} = \frac{1}{16}$.

Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If $k_n = \cos^n\theta + \sin^n\theta$ Then $k_n - k_{n-2} = A\,k_{n-4}$ where A is :-
(A) 1 (B) $\cos^2\theta$
(C) $\sin^2\theta$ (D) $-\sin^2\theta\cos^2\theta$

Q2. The value of $\cos^2 5° + \cos^2 10° + \ldots\ldots + \cos^2 85° + \cos^2 90°$.
(A) 19/2 (B) 17/2 (C) 8 (D) 9

Q3. If $x\cos\theta = y\cos\left(\theta + \frac{2\pi}{3}\right) = z\cos\left(\theta + \frac{4\pi}{3}\right)$ then $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ is equal to :-
(A) 0 (B) 1
(C) 2 (D) None

Q4. Value of expression

$$\frac{\cos 6x+6\cos 4x+15\cos 2x+10}{\cos 5x+5\cos 3x+10\cos x}$$ is :-

(A) cos2x (B) $\cos^2 x$

(C) 2cosx (D) 1 + cosx

Q5. If $\tan\alpha = \frac{1}{7}$, $\sin\beta = \frac{1}{\sqrt{10}}$, then value of $\alpha + 2\beta$ is, and $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Q6. $\cot 5° - \tan 5° - 2\tan 10° - 4\tan 20° - 8\cot 40°$ is equal to :-

(A) 0 (B) 4 tan 40°

(C) 8 tan 40° (D) 8 cot 40°

Q7. If $x \in (\pi, 2\pi)$. Then value of

$$\frac{\sqrt{1+\cos x}+\sqrt{1-\cos x}}{\sqrt{1+\cos x}-\sqrt{1-\cos x}}$$ is -

(A) $\cot\left(\frac{x}{2}+\frac{x}{4}\right)$ (B) $\cot\left(\frac{x}{2}-\frac{x}{4}\right)$

(C) $\cot\left(\frac{\pi}{4}-\frac{x}{2}\right)$ (D) None of these

Q8. Find the exact value of

$$\tan^2\frac{\pi}{16}+\tan^2\frac{3\pi}{16}+\tan^2\frac{5\pi}{16}+\tan^2\frac{7\pi}{16}.$$

(A) 29 (B) 28

(C) 30 (D) 25

Q9. If $f(x) = \frac{\sin 3x}{\sin x}$, $x \neq np$, then the range of values of f(x) for real values of x is-

(A) [–1,3] (B) (–∞,–1]

(C) (3, + ∞) (D) [–1,3)

Q10. The maximum value of expression

$$\frac{1}{\sin^2\theta+3\sin\theta\cos\theta+5\cos^2\theta}$$ is :-

(A) 0 (B) 1

(C) 1/2 (D) 2

Q11. If x and y are real number such that $x^2 + 2xy - y^2 = 6$, find the minimum value of $(x^2 + y^2)^2$.

(A) 20 (B) 18

(C) 17 (D) 15

Q12. Maximum and Minimum value of expression $2\sin^2\theta - 3\sin\theta + 2$ respectively is :-

(A) $\frac{7}{8}, 2$ (B) $\frac{7}{8}, 1$

(C) $\frac{7}{8}, 7$ (D) $7, \frac{7}{8}$

Q13. If $2\sec\alpha - \sin\alpha\tan\alpha \geq p$, $\forall\ a \in [0, \pi/2)$ then p is :-

(A) 2 (B) 3

(C) 4 (D) 5

Q14. Let x = sin1° then value of expression

$$\frac{1}{\cos 0°\cos 1°}+\frac{1}{\cos 1°\cos 2°}+\cdots+\frac{1}{\cos 44°\cos 45°}$$

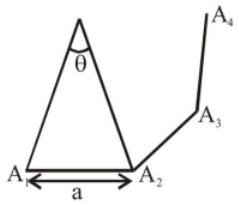
is :-

(A) x (B) $\frac{1}{x}$

(C) $\frac{\sqrt{2}}{x}$ (D) $\frac{x}{\sqrt{2}}$

Q15. If $A_1, A_2, A_3 \ldots\ldots A_n$ are the vertices of a regular n sided polygon inscribed in a circle of radius R.

If $(A_1A_2)^2 + (A_1A_3)^2 + \ldots\ldots + (A_1A_n)^2 = 14R^2$, find the number of sides in the polygon.



(A) 8 (B) 10
(C) 7 (D) 9

Q16. The value of $\sum_{r=0}^{9} \sin^2 \frac{\pi r}{18}$ is equal to-

(A) $\frac{9}{2}$ (B) $\frac{7}{2}$
(C) 5 (D) None of these

Q17. If $\sin A + \sin B = C$ and $\cos A + \cos B = D$ then the value of $\sin(A + B)$ is equal to :-

(A) CD (B) $\frac{CD}{C^2+D^2}$
(C) $\frac{C^2+D^2}{2CD}$ (D) $\frac{2CD}{C^2+D^2}$

Q18. If 'θ' is eliminated from the equations $\cos\theta - \sin\theta = b$ and $\cos3\theta + \sin3\theta = a$, find the eliminant.

(A) $a = 3b - 2b^3$ (B) $a = 3b - 3b^3$
(C) $a = 2b - 2b^3$ (D) None

Exercise - 2A *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If $\cos(A - B) = \frac{3}{5}$ & $\tan A \tan B = 2$, then -

(A) $\cos A \cos B = \frac{1}{5}$

(B) $\sin A \sin B = -\frac{2}{5}$

(C) $\cos(A + B) = -\frac{1}{5}$

(D) $\sin A \sin B = \frac{2}{5}$

Q2. If $\cos(\theta + 10°) = \frac{1}{3}$, then the value of $\tan(\theta + 70°)$ is -

(A) an irrational number

(B) $\frac{(2\sqrt{2}+\sqrt{3})}{(1-2\sqrt{6})}$

(C) $\frac{(2\sqrt{2}-\sqrt{3})}{(1+2\sqrt{6})}$

(D) $\frac{(-2\sqrt{2}+\sqrt{3})}{(1+2\sqrt{6})}$

Q3. If $\sin\theta + \sqrt{\sin\theta + \sqrt{\sin\theta + \sqrt{\sin\theta + ...\infty}}} = \sec^4\alpha$, then $\sin\theta$ is equal to -

(A) $\sec^2\alpha \tan^2\alpha$ (B) $2\frac{(1-\cos 2\alpha)}{(1+\cos 2\alpha)^2}$
(C) $2\frac{(1+\cos 2\alpha)}{(1-\cos 2\alpha)^2}$ (D) $\cot^2\alpha \operatorname{cosec}^2\alpha$

Q4. Which of the following when simplified reduces to unity? (whenever defined)

(A) $\dfrac{1-2\sin^2\alpha}{2\cot\left(\frac{\pi}{4}+\alpha\right)\cos^2\left(\frac{\pi}{4}-\alpha\right)}$

(B) $\dfrac{\sin(\pi-\alpha)}{\sin\alpha-\cos\alpha\tan\frac{\alpha}{2}}+\cos(\pi-\alpha)$

(C) $\dfrac{1}{4\sin^2\alpha\cos^2\alpha}+\dfrac{(1-\tan^2\alpha)^2}{4\tan^2\alpha}$

(D) $\dfrac{1+\sin 2\alpha}{(\sin\alpha+\cos\alpha)^2}$

Q5. Factors of $\cos4\theta - \cos4\phi$ are -

(A) $(\cos\theta + \cos\phi)$ (B) $(\cos\theta - \cos\phi)$

(C) $(\cos\theta + \sin\phi)$ (D) $(\cos\theta - \sin\phi)$

Q6. If $\dfrac{1+\cos 2x}{\sin 2x}+3\left(1+(\tan x)\tan\dfrac{x}{2}\right)\sin x = 4$, then the value of tanx can be equal to -

(A) 1 (B) $\frac{1}{2}$ (C) 3 (D) $\frac{1}{3}$

Q7. Which of the following hold(s) good ?

(A) $\tan(45°) - \tan20° - \tan25° > 0$

(B) $\sin20° + \sin40° - \cos10° < 0$

(C) If $\sin x = 3\cos x$, then

$$\frac{(\tan 2x)(\sin 2x)}{\cos^2 x} < 0$$

(D) $\tan^2\left(\dfrac{\pi}{24}\right)+\cot^2\left(\dfrac{\pi}{24}\right) > 55$

Q8. In a ΔABC,

Let $\sqrt{\sin A}+\sqrt{\sin B}+\sqrt{\sin C}$

$$=\sqrt{12\cos\frac{A}{2}.\cos\frac{B}{2},\cos\frac{C}{2}}$$

then which of the following option is / are correct.

(A) $\dfrac{\sin^2 A}{\sin^2 B}+\dfrac{\sin^2 B}{\sin^2 C}+\dfrac{\sin^2 C}{\sin^2 A}=3$

(B) ΔABC will be equilateral

(C) ΔABC will right angle

(D) $\dfrac{\sin^2 A}{\sin^2 B}+\dfrac{\sin^2 B}{\sin^2 C}+\dfrac{\sin^2 C}{\sin^2 A}=2$

Q9. If $(3 - 4\sin^2 1)(3 - 4\sin^2 3)(3 - 4\sin^2 3^2)$ ..... $(3 - 4\sin^2(3^{n-1})) = \sin a/\sin b$, where $n \in N$ & a, b are integers in radian, then the digit at the unit place of (a + b) may be-

(A) 4 (B) 0

(C) 8 (D) 2

Q10. For a positive integer n, let

$$f_n(\theta)=\left(\tan\frac{\theta}{2}\right)(1+\sec\theta)(1+\sec 2\theta)$$

$$(1+\sec 4\theta)....(1+\sec 2^n\theta).$$

Then

(A) $f_2\left(\dfrac{\pi}{16}\right)=1$ (B) $f_3\left(\dfrac{\pi}{32}\right)=1$

(C) $f_4\left(\dfrac{\pi}{64}\right)=1$ (D) $f_5\left(\dfrac{\pi}{128}\right)=1$

Q11. If $2\tan10° + \tan50° = 2x$,

$\tan20° + \tan50° = 2y$,

$2\tan10° + \tan70° = 2w$

and $\tan20° + \tan70° = 2z$,

then which of the following is/are true -

(A) $z > w > y > x$

(B) $w = x + y$

(C) $2y = z$

(D) $z + x = w + y$

Q12. If $\sec\theta + \tan\theta = 1$, then one root of equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ is -

(A) $\tan\theta$ (B) $\sec\theta$

(C) $\cos\theta$ (D) $\sin\theta$

Q13. If $\frac{\tan 3A}{\tan A} = k, (k \neq 1)$, then -

(A) $\frac{\cos A}{\cos 3A} = \frac{k-1}{2}$ (B) $\frac{\sin 3A}{\sin A} = \frac{2k}{k-1}$

(C) $k < \frac{1}{3}$ (D) $k > 3$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 14 to 15)

Let $\sin\alpha + \sin\beta = \frac{\sqrt{6}}{3}$ and $\cos\alpha + \cos\beta = \frac{\sqrt{3}}{3}$.

Q14. The value of $\tan\left(\frac{\alpha+\beta}{2}\right)$ is equal to

(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\sqrt{3}$ (D) $\frac{1}{\sqrt{3}}$

Q15. The value of $100\cos^2\left(\frac{\alpha-\beta}{2}\right)$ is equal to -

(A) 5 (B) 25

(C) 50 (D) 75

Paragraph # 2 (Ques. 16 to 17)

For $n \in N$, let $f(n) = 8(\cos^n\theta - \sin^n\theta)$
and $g(n) = 8(\cos^n\theta + \sin^n\theta)$

Q16. If $f(6) = a\cos(2\theta) + b\cos(6\theta), \forall\ \theta \in R$ then the value of $(a + b)$ is equal to -

(A) 8 (B) 7

(C) 6 (D) 5

Q17. The value of $\left(\frac{g(6)}{32} + \frac{3}{16}\sin^2 2\theta\right)$ is equal to -

(A) 0 (B) $\frac{1}{4}$ (C) $\frac{-1}{4}$ (D) -2

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q18. $\cos A + \cos B = a$, $\sin A + \sin B = b$

Column-I	**Column-II**
(A) $\tan\frac{A+B}{2}$	(P) $\frac{2ab}{a^2+b^2}$
(B) $\sin(A + B)$	(Q) b/a
(C) $\cos(A - B)$	(R) $\frac{a^2-b^2}{a^2+b^2}$
(D) $\cos(A + B)$	(S) $\frac{a^2+b^2-2}{2}$

SUBJECTIVE TYPE

Q19. If $(1 + \sin t)(1 + \cos t) = \frac{5}{4}$. Find the value $(1 - \sin t)(1 - \cos t)$. $[\sqrt{10} = 3.17]$

Q20. Given that $3\sin x + 4\cos x = 5$, where $x \in \left(0, \frac{\pi}{2}\right)$.

Find the value of $2\sin x + \cos x + 4\tan x$.

Q21. Prove that : $\cos^2\alpha + \cos^2(\alpha + \beta) - 2\cos\alpha\cos\beta\cos(\alpha + \beta) = \sin^2\beta$

Q22. Prove that : $\cos 2\alpha = 2\sin^2\beta + 4\cos(\alpha+\beta)\sin\alpha\ \sin\beta + \cos 2(\alpha+\beta)$

Q23. Prove that : $\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha = \cot\alpha$.

Q24. If $(4\cos^2 40° - 3)(3 - 4\sin^2 40°) = a + b\cos 20°$, find $(a - b)$.

Q25. Prove that :

(i) $\tan 20° \cdot \tan 40° \cdot \tan 60° \cdot \tan 80° = 3$

(ii) $\tan 9° - \tan 27° - \tan 63° + \tan 81° = 4$.

(iii) $\sin^4\frac{\pi}{16} + \sin^4\frac{3\pi}{16} + \sin^4\frac{5\pi}{16} + \sin^4\frac{7\pi}{16} = \frac{3}{2}$

Q26. Calculate without using trigonometric tables:

(i) $4\cos 20° - \sqrt{3}\cot 20°$

(ii) $\dfrac{2\cos 40° - \cos 20°}{\sin 20°}$

(iii) $\cos^6\frac{\pi}{16} + \cos^6\frac{3\pi}{16} + \cos^6\frac{5\pi}{16} + \cos^6\frac{7\pi}{16}$

(iv) $\tan 10° - \tan 50° + \tan 70°$

Q27. If $f(\theta) = \sum_{n=1}^{6} \operatorname{cosec}\left(\theta + \frac{(n-1)\pi}{4}\right)\operatorname{cosec}\left(\theta + \frac{n\pi}{4}\right)$, where $0 < \theta < \frac{\pi}{2}$, then find the minimum value of $f(\theta)$.

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
B	C	C	A	B	C	C	D	A	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	B	B	C	A	A	B	C	A	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	B	D	D	D	D	A	C	A	C
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
D	A	B	B	C	C	D	C	A	C
41	**42**	**43**							
B	D	B							

Ex - 1A

1. (A,D) **2.** (B,C,D) **3.** (B,C) **4.** (A,C) **5.** (A,B)

6. (B) **7.** (D) **8.** (D) **9.** (A) **10.** (D)

11. (A) → (Q); (B) → (P);(C) → (S); (D) → (R) **12.** 22.28

13. 120°: II[nd] quad; 750°: I[st] quad; –300°: I[st] quad **14.** 5 : 4

18. (i) 0, (ii) $-\frac{1}{\sqrt{3}}$, (iii) $-\cos\theta$, (iv) $\frac{1}{2}$ **19.** $\left(\frac{7\pi}{6}\text{ or }\frac{11\pi}{6}\right)$ **21.** negative

23. $\frac{1}{81}$ **24.** 1 **29.** 4 **30.** –1 **31.** 4

32. $-\frac{1}{2}$ **37.** $8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

38. (i) $y \in [-1, 1]$, (ii) $y \in [0, 1]$, (iii) $y \in [-1, 1]$ **39.** $\left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$

40. min: –4, max: +4 **41.** (i) [–13, 13] , (ii) [–6, 20]

42. (i) [24] , (ii) [49] **43.** 31 **44.** $y_{min} = 26$

Ex - 2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	B	A	C	D	A	A	B	D	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**		
B	D	A	B	C	C	D	A		

Ex - 2A

1. A,C,D **2.** A,B,D **3.** A,B **4.** (A,B,D) **5.** (A,B,C,D)

6. (A,D) **7.** (A,C,D) **8.** (A,B) **9.** (A,B,C,D) **10.** (A,B,C,D)

11. (A,B,C,D) **12.** (B,C) **13.** (A,B,C,D) **14.** (A) **15.** (B)

16. (A) **17.** (B) **18.** (A) → q; (B) → r ; (C) → s; (D) → p

19. 6.42 or $\frac{13}{4}-\sqrt{10}$ **20. 5** **24.** 3

26. (i) –1 (ii) $\sqrt{3}$ (iii) $\frac{5}{4}$ (iv) $\sqrt{3}$ **27.** $2\sqrt{2}$

-: Important Notes & Formulas :-

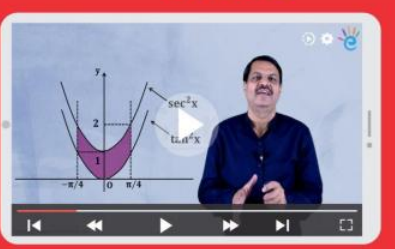
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